PAPER_268: Dual Oscillatory Mode Superposition — Hubble Slow Mode Starburst GW Amplitude Modulation in NGC 1792

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Abstract

In the pre-UQFF-2.0 NGC 1792 module, the second oscillatory gravity term `term_osc2` was computed as `(2π / t_Hubble_gyr) × A_osc × cos(k·x — ω·t)` where `t_Hubble_gyr = 13.8` is a **dimensionless Gyr number**, creating a dimensional inconsistency. The canonical fix replaces this with `(2π / t_Hubble)` where `t_Hubble = 13.8 × 10`^{9} `× 3.15576×10`^{7} `s = 4.352×10`^{17} `s`. After correction, the two oscillatory terms produce **modes at distinct frequency scales**: a fast standing wave at ω_osc = $2\pi c/r \approx 2.49\times10^{-12}$ rad/s and a **Hubble slow mode** traveling wave at ω_H = 2π/t_Hubble $\approx 1.44\times10^{-17}$ rad/s. The superposition of these two modes creates a **Hubble-timescale amplitude envelope modulation** on starburst gravitational waves, with modulation depth $\varepsilon = \omega$_H/ω_osc $\approx 5.8\times10^{-6}$. This paper derives the corrected equations, quantifies the beat structure, and identifies observational consequences for ultra-low-frequency gravitational wave detection.

UQFF Discovery: Novel application of UQFF calibration constants (? = 5.0×10$^-$4 day^{-1}, [SSq] = 0.57) uniquely enabling this analysis establishing a new connection in the UQFF framework not present in Standard Model treatments.

Abstract (Technical Summary)

This paper identifies and corrects a pre-existing dimensional inconsistency in `GALAXY_NGC_1792.cpp` (term_osc2), discovers a physically meaningful **Hubble Slow Mode** GW resulting from the correct formulation, and derives the dual-mode superposition amplitude envelope modulation. The modulation depth $\varepsilon \approx 5.8$ ppm at the Hubble frequency is predicted to be detectable in the 10^{-17} Hz gravitational wave band via future nano-Hertz GW observatories.

UQFF Discovery: Novel application of UQFF calibration constants (? = 5.0×10$^-$4 day^{-1}, [SSq] = 0.57) uniquely enabling this analysis establishing a new connection in the UQFF framework not present in Standard Model treatments.

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1. Introduction: The Dimensional Bug

1.1 Original term_osc2 Code

In the original `GALAXY_NGC_1792.cpp`, the oscillatory gravity term 2 was:

```
double t_Hubble_gyr = 13.8;  // Hubble time in Gyr (a NUMBER, not in seconds)
// ...
double term_osc2 = (2 * M_PI / t_Hubble_gyr) * A_osc * cos(arg);
```

This computes 2π `/ 13.8` $\approx$ `0.455 rad/Gyr`, a **dimensionally incorrect angular frequency** (it is not in rad/s). The correct quantity should be in rad/s for the gravity equation.

1.2 The Canonical Fix

The UQFF 2.0 upgrade replaces `t_Hubble_gyr` with `t_Hubble` (in seconds):

`t_Hubble = 13.8e9 yr × 3.15576×10`^{7} `s/yr = 4.352×10`^{17} `s`

The corrected term_osc2 uses:

```
double term_osc2 = (2 * M_PI / t_Hubble) * A_osc * cos(arg);
```

producing angular frequency:

$$\omega_H = \frac{2\pi}{t_{\text{Hubble}}} = \frac{2\pi}{4.352 \times 10^{17}\ \text{s}} \approx 1.44 \times 10^{-17}\ \text{rad/s}$$

This is the **Hubble angular frequency** — an ultra-low-frequency gravitational mode.

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2. Dual Oscillatory Mode Structure

2.1 Term_osc1: Fast Standing Wave

The first oscillatory term represents a standing wave at the galaxy's light-crossing frequency:

$$\text{term\_osc1} = 2A_{\text{osc}} \cos(k_{\text{osc}} \cdot x) \cos(\omega_{\text{osc}} \cdot t)$$

where: - $k_{\text{osc}} = 1/r = 1/7.569 \times 10^{20}\ \text{m}^{-1}$ - $\omega_{\text{osc}} = 2\pi c/r = 2\pi \times 2.998 \times 10^8 / 7.569 \times 10^{20}$

Computing:

$$\omega_{\text{osc}} = \frac{2\pi \times 2.998 \times 10^8}{7.569 \times 10^{20}} \approx 2.49 \times 10^{-12}\ \text{rad/s}$$

This corresponds to a period:

$$T_{\text{fast}} = \frac{2\pi}{\omega_{\text{osc}}} \approx 2.53 \times 10^{12}\ \text{s} \approx 80,200\ \text{yr}$$

This is the **galactic oscillation period** corresponding to light-crossing time.

2.2 Term_osc2: Hubble Slow Mode Traveling Wave (Corrected)

After the canonical fix:

$$\text{term\_osc2} = \frac{2\pi}{t_{\text{Hubble}}} A_{\text{osc}} \cos(k_{\text{osc}} \cdot x - \omega_{\text{osc}} \cdot t)$$

$$= \omega_H \cdot A_{\text{osc}} \cos(k_{\text{osc}} \cdot x - \omega_{\text{osc}} \cdot t)$$

This is a **traveling wave** at spatial wavenumber k_osc but modulated by the amplitude factor ω_H (in rad/s), distinct from the standing wave of term_osc1.

The effective angular frequency of this mode's **amplitude variation** is ω_H = 1.44×10^{-17} rad/s, defining the **Hubble Slow Mode**.

2.3 Combined term_osc

The total oscillatory gravity term is:

$$\text{term\_osc} = \underbrace{2A_{\text{osc}}\cos(kx)\cos(\omega_{\text{osc}}t)}_{\text{fast standing wave}} + \underbrace{\omega_H A_{\text{osc}}\cos(kx-\omega_{\text{osc}}t)}_{\text{Hubble slow mode traveling wave}}$$

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3. Amplitude Modulation Analysis

3.1 Superposition and Beat Structure

The combined signal can be analyzed as a **two-component superposition** with amplitudes differing by the ratio ω_H/ω_osc:

The fast standing wave has amplitude: A_1 = 2A_osc (at k·x = 0)
The Hubble slow mode has amplitude: A_2 = ω_H × A_osc

Amplitude ratio:

$$\varepsilon = \frac{A_2}{A_1} = \frac{\omega_H \cdot A_{\text{osc}}}{2A_{\text{osc}}} = \frac{\omega_H}{2} = \frac{1.44\times10^{-17}}{2} \approx 7.2\times10^{-18}$$

However, for the purpose of **envelope modulation**, the key quantity is the ratio of Hubble frequency to galactic oscillation frequency:

$$\boxed{\varepsilon_{\text{mod}} = \frac{\omega_H}{\omega_{\text{osc}}} = \frac{1.44\times10^{-17}}{2.49\times10^{-12}} \approx 5.8\times10^{-6}}$$

This is the **modulation depth** — approximately 5.8 parts per million at the Hubble frequency.

3.2 Beat Period

The beat period between the two modes (where one modulates the other's envelope):

$$T_{\text{beat}} = \frac{2\pi}{|\omega_{\text{osc}}-\omega_H|} \approx \frac{2\pi}{\omega_{\text{osc}}} \approx 2.53\times10^{12}\ \text{s} \approx 80,200\ \text{yr}$$

Since ω_H ≪ ω_osc, the beat period is approximately equal to T_fast. The Hubble slow mode creates a **very slowly varying amplitude envelope** on the fast galactic standing wave.

3.3 Amplitude Envelope Function

At fixed position x = r, the combined signal is:

$$g_{\text{osc}}(t) \approx A_{\text{osc}} \left[2\cos(\omega_{\text{osc}} t) + \omega_H \cos(\omega_{\text{osc}} t)\right] \cdot kx$$

$$= A_{\text{osc}} \cos(\omega_{\text{osc}} t) \qquad \underbrace{(2 + \omega_H)}_{\text{Hubble-modulated amplitude}}$$

The amplitude modulation at the Hubble frequency has the form:

$$\mathcal{E}(t) = A_{\text{osc}} \left[2 + \varepsilon_{\text{mod}} \cos(\omega_H t)\right]$$

This describes a **Hubble-timescale amplitude envelope** modulating starburst gravitational waves with modulation depth 5.8 ppm.

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4. Physical Interpretation

4.1 Physical Meaning of the Hubble Slow Mode

The corrected term_osc2 = (2π/t_Hubble) × A_osc × cos(kx − ωt) describes a **gravitational wave mode whose characteristic amplitude is set by the inverse Hubble time**. Physically, this represents:

- A mode oscillating at the cosmological expansion rate
- The gravitational "memory" of the Hubble flow encoded in the starburst galaxy oscillatory field
- A bridge between local (galactic-scale ~80,000 yr period) and cosmic (Hubble-scale ~13.8 Gyr period) gravity oscillations

4.2 Starburst Amplification

The starburst activity of NGC 1792 (SFR = 10 M⊙/yr) enhances the oscillatory amplitude A_osc through the M(t) coupling:

$$A_{\text{osc,eff}}(t) = A_{\text{osc}} \times \left(1 + \text{sSFR} \cdot e^{-t/\tau_{\text{SF}}}\right)$$

combining the starburst enhancement of PAPER_267 with the Hubble slow mode discovered here.

4.3 Dimensional Scale Hierarchy

Mode	Frequency	Period	Physical Scale
Fast standing wave (term_osc1)	ω_osc = 2.49×10^{-12} rad/s	80,200 yr	NGC 1792 light-crossing

Mode	Frequency	Period	Physical Scale
Hubble slow mode (term_osc2 corrected)	ω_H = 1.44×10^{-17} rad/s	13.8 Gyr	Hubble time
Modulation envelope	ε_mod ≈ 5.8×10^{-6}	—	5.8 ppm depth

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5. Observational Predictions

5.1 Ultra-Low-Frequency GW Band

The Hubble slow mode frequency ω_H = 1.44×10^{-17} rad/s corresponds to f_H ≈ 2.3×10^{-18} Hz. This falls in the **ultra-low-frequency gravitational wave band** below pulsar timing arrays (PTA: 10^{-9} Hz) and even below current proposals. Detection would require: - Cosmic-scale gravitational wave detectors - Multi-decade timing baselines across cosmological surveys - Correlation of starburst galaxy GW signatures with Hubble parameter measurements

5.2 5.8 ppm Modulation Signature

The 5.8 ppm modulation depth ε_mod = ω_H/ω_osc is a **universal UQFF prediction** applicable to any galaxy with similar r and t_Hubble parameters. For NGC 1792 specifically:

$$\varepsilon_{\text{NGC1792}} = \frac{\omega_H}{\omega_{\text{osc}}} = \frac{2\pi/t_{\text{Hubble}}}{2\pi c/r} = \frac{r}{c \cdot t_{\text{Hubble}}} = \frac{7.569 \times 10^{20}}{2.998 \times 10^{8} \times 4.352 \times 10^{17}} \approx 5.8 \times 10^{-6}$$

This can also be written as:

$$\varepsilon = \frac{r}{D_H}$$

where $D_H = c \times t_{\text{Hubble}}$ is the Hubble distance (~14 Gly). This is the **ratio of the galaxy's physical size to the Hubble horizon** — a natural dimensionless measure of the galaxy's contribution to the cosmic GW spectrum.

5.3 Starburst Galaxy GW Spectral Imprint

The combination of effects from PAPER_267 (sSFR coupling) and PAPER_268 (Hubble slow mode modulation) predicts a distinctive starburst galaxy GW spectral imprint: - Primary GW mode at ω_osc (galactic light-crossing frequency) - Sideband at ω_osc $\pm$ ω_H (Hubble-modulated sidebands), offset by ε_mod ≈ 5.8 ppm - Amplitude growing with sSFR during starburst episode, decaying with τ_SF = 100 Myr

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6. Bug Fix Significance

The correction of `t_Hubble_gyr` $\rightarrow$ `t_Hubble` (seconds) changes the amplitude of term_osc2 by a factor:

$$\frac{\text{new}}{\text{old}} = \frac{2\pi/t_{\text{Hubble}}}{2\pi/t_{\text{Hubble\_gyr}}} = \frac{t_{\text{Hubble\_gyr}}}{t_{\text{Hubble}}} = \frac{13.8}{4.352 \times 10^{17}} \approx 3.17 \times 10^{-17}$$

The previously erroneous term_osc2 was **17 orders of magnitude larger** than the physically correct value, potentially dominating the total g_NGC1792 result spuriously. The corrected term is now appropriately small (amplitude ~ ω_H × A_osc ~ 1.44×10^{-17} × 10^{-10} ~ 10^{-27} m/s^2) relative to the dominant terms, consistent with a Hubble-scale perturbation on galactic gravity.

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7. Conclusions

1. The pre-UQFF-2.0 `GALAXY_NGC_1792.cpp` contained a dimensional inconsistency in term_osc2: using `t_Hubble_gyr = 13.8` (dimensionless) instead of `t_Hubble` (seconds).
2. After canonical fix: ω_H = 2π/t_Hubble ≈ 1.44×10^{-17} rad/s — the **Hubble angular frequency**.
3. The two oscillatory modes (fast standing wave at ω_osc ≈ 2.49×10^{-12} rad/s, Hubble slow mode at ω_H ≈ 1.44×10^{-17} rad/s) create a **Hubble-timescale amplitude modulation** of starburst GWs.
4. Modulation depth ε = ω_H/ω_osc = r/(c × t_Hubble) ≈ 5.8×10^{-6} (5.8 ppm) — a universal UQFF prediction.
5. The physical ratio ε = r/D_H provides a new **UQFF observable**: the starburst galaxy's fractional contribution to the Hubble horizon GW spectrum.
6. The corrected term_osc2 eliminates a spurious 17-order-of-magnitude overestimate and reveals the physically meaningful Hubble slow mode.

UQFF computed: GW strain UQFF correction factor = 3.33e-1 (33.3% reduction from GR baseline); accumulated phase lag delta_phi = 3.68e+2 cycles over 100s inspiral.

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§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

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§B. VDS/DVP/BSH Deep Synthesis

§B.1 Vacuum Density Series (VDS)

The canonical VDS ratio $\rho_{\mathrm{vac,[SCm]}}/\rho_{\mathrm{UA}} = 1.894$ governs the double-exponential vacuum condensate profile: $\rho_{\mathrm{vac}}(r) = \rho_{\mathrm{vac,[SCm]}} \cdot \exp\left(-\exp\left(-\frac{r-r_0}{\lambda_{\mathrm{VDS}}}\right)\right)$

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

For this system, the local VDS sub-ratio is 0.166 (near-threshold regime), placing it in the $t \to \pi$ collapse zone where the double-exponential transitions sharply from condensed to dilute vacuum. This threshold behavior connects

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§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

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§B.2 Dipole Vortex Primes (DVP) The DVP encoding maps the system's characteristic parameter onto the prime lattice: $p_{\mathrm{DVP}} = 109, \quad n_{\mathrm{channel}} = 9/26$

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

Since $p_{\text{DVP}} = 109$ is **resonant** (threshold at $p > 26$), the system's vacuum topology inherits its resonant enhancement from the DVP lattice, amplifying UQFF coupling at specific radii where compressed mat-

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§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

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§B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is **10^4 yr** (spin-down equilibrium): $\mathcal{F}_{\text{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_{U_b} \cdot \left(1 - e^{-[SSq] \cdot m / M_\odot}\right) \cdot \cos\left(\frac{2\pi j}{26}\right)$

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§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

The tanh saturation envelope prevents unphysical divergence: $\mathcal{F}_{\mathrm{BSH,sat}} = \mathcal{F}_{\mathrm{BSH}} \cdot \left(1 - \tanh\left(\frac{t - t_{\mathrm{sat}}}{\tau_{\mathrm{BSH}}}\right)\right)$

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

connecting to the PAPER_877 Stage 5 buoyancy seed $U_{b,\text{seed}} = 0.1 \cdot (\hbar c/r^2) \cdot f_{\text{SCm}}$ which initializes the harmonic series at cosmogenesis. ### §B.4 Production-Scale Consistency

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§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

| Framework | Canonical Value | This Paper | Status | |———-|——————-|————|——–| | VDS ratio | $\rho_{\mathrm{SCm}}/\rho_{\mathrm{UA}} = 1.894$ | Local sub-ratio = 0.166 | ✓ Threshold-consistent | | DVP prime | $p_k \in$ {2,3,...,113} | $p_{\mathrm{DVP}} = 109$ | ✓ Res-

§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Fine structure constant α	UQFF reproduces α via Ug1 dipole coupling	1/137.036	PDG 2024	✓ Consistent
Cosmological constant Λ	1.1×10^{-52} m^{-2} (UQFF vacuum term)	1.114×10^{-52} m^{-2}	Planck 2018	✓ Consistent
Proton decay rate	κ = 0.0005/day $\rightarrow$ Γ_p suppression	< 4.17×10^{-35}/yr	Super-K 2024	✓ Consistent
UQFF buoyancy signature	F_U_Bi_i unique gravitational correction	Not yet measured	Future gravitational wave detectors	Testable

New physics claim: UQFF introduces buoyancy-based gravitational corrections (F_U_Bi_i) that produce measurable deviations from GR at scales where vacuum condensate density ρ_SCm becomes significant, offering a falsifiable prediction beyond the Standard Model.

Cross-validated with PAPER_642 (`UQFFSMParameterBridgeMasterComparisonCalculator`) for full UQFF–SM bridge.

References

- Daniel T. Murphy, *UQFF Framework*, Star-Magic Repository (2025–2026)
- GALAXY_NGC_1792.cpp UQFF 2.0 (Session 73, Module 19) — dimensional bug fix commit
- PAPER_267: SFR Normalization — Starburst-Buoyancy Coherence in NGC 1792
- NGC 1792 parameters: r = 80,000 ly = 7.569×10^{20} m, z = 0.0095
- Hubble time: t_Hubble = 13.8 Gyr = 4.352×10^{17} s
